Attention Scaffolding for Sustained Long-Form Reading

A. Reader, B. Writer, and C. Editor

IEEE Transactions on Learning Technologies, vol. 14, no. 3, 2021
Digital Object Identifier 10.1109/TLT.2021.1234567

Abstract-Long-form reading has declined among undergraduates while short-form media consumption has risen. We present a reading platform that borrows retention mechanics from streaming interfaces and applies them to book reading. In a three-week study with twenty participants we measure session length, comprehension retention and self-reported engagement. Sessions lengthened by a mean of four minutes and comprehension scores did not decline, suggesting that habit scaffolding need not come at the cost of depth.

I. INTRODUCTION

The activation energy required to begin a reading session is disproportionately high when compared with low-effort alternatives. Prior work attributes this to the absence of automated progression rather than to any deficit in reading ability. We take the position that modern behavioural mechanics should be repurposed rather than resisted.

The activation energy required to begin a reading session is disproportionately high when compared with low-effort alternatives. Prior work attributes this to the absence of automated progression rather than to any deficit in reading ability. We take the position that modern behavioural mechanics should be repurposed rather than resisted.

The activation energy required to begin a reading session is disproportionately high when compared with low-effort alternatives. Prior work attributes this to the absence of automated progression rather than to any deficit in reading ability. We take the position that modern behavioural mechanics should be repurposed rather than resisted.

The activation energy required to begin a reading session is disproportionately high when compared with low-effort alternatives. Prior work attributes this to the absence of automated progression rather than to any deficit in reading ability. We take the position that modern behavioural mechanics should be repurposed rather than resisted.

The activation energy required to begin a reading session is disproportionately high when compared with low-effort alternatives. Prior work attributes this to the absence of automated progression rather than to any deficit in reading ability. We take the position that modern behavioural mechanics should be repurposed rather than resisted.

The activation energy required to begin a reading session is disproportionately high when compared with low-effort alternatives. Prior work attributes this to the absence of automated progression rather than to any deficit in reading ability. We take the position that modern behavioural mechanics should be repurposed rather than resisted.

The activation energy required to begin a reading session is disproportionately high when compared with low-effort alternatives. Prior work attributes this to the absence of automated progression rather than to any deficit in reading ability. We take the position that modern behavioural mechanics should be repurposed rather than resisted.

Our contribution is threefold. First, we describe a segmentation method that cuts a book at narrative boundaries. Second, we report a pilot deployment. Third, we release the instrumentation schema so that other groups can replicate the measurement.

Our contribution is threefold. First, we describe a segmentation method that cuts a book at narrative boundaries. Second, we report a pilot deployment. Third, we release the instrumentation schema so that other groups can replicate the measurement.

Our contribution is threefold. First, we describe a segmentation method that cuts a book at narrative boundaries. Second, we report a pilot deployment. Third, we release the instrumentation schema so that other groups can replicate the measurement.

Our contribution is threefold. First, we describe a segmentation method that cuts a book at narrative boundaries. Second, we report a pilot deployment. Third, we release the instrumentation schema so that other groups can replicate the measurement.

Our contribution is threefold. First, we describe a segmentation method that cuts a book at narrative boundaries. Second, we report a pilot deployment. Third, we release the instrumentation schema so that other groups can replicate the measurement.

Our contribution is threefold. First, we describe a segmentation method that cuts a book at narrative boundaries. Second, we report a pilot deployment. Third, we release the instrumentation schema so that other groups can replicate the measurement.

II. RELATED WORK

Dual coding theory holds that symbolic and imagistic representations are encoded along separate pathways. The original formulation by Paivio has been extended to reading comprehension by a number of authors. Separately, the literature on variable reward scheduling describes why intermittent reinforcement sustains engagement more effectively than fixed schedules.

Dual coding theory holds that symbolic and imagistic representations are encoded along separate pathways. The original formulation by Paivio has been extended

to reading comprehension by a number of authors. Separately, the literature on variable reward scheduling describes why intermittent reinforcement sustains engagement more effectively than fixed schedules.

Dual coding theory holds that symbolic and imagistic representations are encoded along separate pathways. The original formulation by Paivio has been extended to reading comprehension by a number of authors. Separately, the literature on variable reward scheduling describes why intermittent reinforcement sustains engagement more effectively than fixed schedules.

Dual coding theory holds that symbolic and imagistic representations are encoded along separate pathways. The original formulation by Paivio has been extended to reading comprehension by a number of authors. Separately, the literature on variable reward scheduling describes why intermittent reinforcement sustains engagement more effectively than fixed schedules.

Dual coding theory holds that symbolic and imagistic representations are encoded along separate pathways. The original formulation by Paivio has been extended to reading comprehension by a number of authors. Separately, the literature on variable reward scheduling describes why intermittent reinforcement sustains engagement more effectively than fixed schedules.

Dual coding theory holds that symbolic and imagistic representations are encoded along separate pathways. The original formulation by Paivio has been extended to reading comprehension by a number of authors. Separately, the literature on variable reward scheduling describes why intermittent reinforcement sustains engagement more effectively than fixed schedules.

III. METHOD

Participants were recruited from an undergraduate population. Each was assigned one of five public-domain novels. The platform logged every user-facing interaction with a timestamp, and active reading time excluded intervals during which the browser tab was hidden.

Participants were recruited from an undergraduate population. Each was assigned one of five public-domain novels. The platform logged every user-facing interaction with a timestamp, and active reading time excluded intervals during which the browser tab was hidden.

Participants were recruited from an undergraduate population. Each was assigned one of five public-domain novels. The platform logged every user-facing interaction with a timestamp, and active reading time excluded intervals during which the browser tab was hidden.

Participants were recruited from an undergraduate population. Each was assigned one of five public-domain novels. The platform logged every user-facing interaction with a timestamp, and active reading time excluded intervals during which the browser tab was hidden.

Participants were recruited from an undergraduate population. Each was assigned one of five public-domain novels. The platform logged every user-facing interaction with a timestamp, and active reading time excluded intervals during which the browser tab was hidden.

Segment length was held between four hundred and six hundred words. The boundary detector preferred positions at which a paragraph ended with an unresolved question, on the assumption that such positions function as cliffhangers.

Segment length was held between four hundred and six hundred words. The boundary detector preferred positions at which a paragraph ended with an unresolved question, on the assumption that such positions function as cliffhangers.

Segment length was held between four hundred and six hundred words. The boundary detector preferred positions at which a paragraph ended with an unresolved question, on the assumption that such positions function as cliffhangers.

Segment length was held between four hundred and six hundred words. The boundary detector preferred positions at which a paragraph ended with an unresolved question, on the assumption that such positions function as cliffhangers.

Segment length was held between four hundred and six hundred words. The boundary detector preferred positions at which a paragraph ended with an unresolved question, on the assumption that such positions function as cliffhangers.

Segment length was held between four hundred and six hundred words. The boundary detector preferred positions at which a paragraph ended with an unresolved question, on the assumption that such positions function as cliffhangers.

Segment length was held between four hundred and six hundred words. The boundary detector preferred positions at which a paragraph ended with an unresolved question, on the assumption that such

positions function as cliffhangers.

IV. RESULTS

Mean session length rose from eleven to fifteen minutes over the study period. Comprehension checkpoint scores were flat, with a mean of 0.72 in week one and 0.74 in week three. Eighteen of twenty participants reported that the recap feature reduced the friction of resuming a book.

Mean session length rose from eleven to fifteen minutes over the study period. Comprehension checkpoint scores were flat, with a mean of 0.72 in week one and 0.74 in week three. Eighteen of twenty participants reported that the recap feature reduced the friction of resuming a book.

Mean session length rose from eleven to fifteen minutes over the study period. Comprehension checkpoint scores were flat, with a mean of 0.72 in week one and 0.74 in week three. Eighteen of twenty participants reported that the recap feature reduced the friction of resuming a book.

Mean session length rose from eleven to fifteen minutes over the study period. Comprehension checkpoint scores were flat, with a mean of 0.72 in week one and 0.74 in week three. Eighteen of twenty participants reported that the recap feature reduced the friction of resuming a book.

Mean session length rose from eleven to fifteen minutes over the study period. Comprehension checkpoint scores were flat, with a mean of 0.72 in week one and 0.74 in week three. Eighteen of twenty participants reported that the recap feature reduced the friction of resuming a book.

Mean session length rose from eleven to fifteen minutes over the study period. Comprehension checkpoint scores were flat, with a mean of 0.72 in week one and 0.74 in week three. Eighteen of twenty participants reported that the recap feature reduced the friction of resuming a book.

Mean session length rose from eleven to fifteen minutes over the study period. Comprehension checkpoint scores were flat, with a mean of 0.72 in week one and 0.74 in week three. Eighteen of twenty participants reported that the recap feature reduced the friction of resuming a book.

V. DISCUSSION

The absence of a comprehension penalty is the result we consider most load-bearing. A platform that lengthens sessions by degrading attention quality would not be worth deploying. We note that our sample is small and drawn from a single institution.

The absence of a comprehension penalty is the result we consider most load-bearing. A platform that lengthens sessions by degrading attention quality would not be worth deploying. We note that our sample is small and drawn from a single institution.

The absence of a comprehension penalty is the result we consider most load-bearing. A platform that lengthens sessions by degrading attention quality would not be worth deploying. We note that our sample is small and drawn from a single institution.

The absence of a comprehension penalty is the result we consider most load-bearing. A platform that lengthens sessions by degrading attention quality would not be worth deploying. We note that our sample is small and drawn from a single institution.

The absence of a comprehension penalty is the result we consider most load-bearing. A platform that lengthens sessions by degrading attention quality would not be worth deploying. We note that our sample is small and drawn from a single institution.

VI. CONCLUSION

Habit scaffolding borrowed from streaming interfaces lengthened reading sessions without measurable comprehension cost. Future work should test whether revealing artwork suppresses self-generated imagery.

Habit scaffolding borrowed from streaming interfaces lengthened reading sessions without measurable comprehension cost. Future work should test whether revealing artwork suppresses self-generated imagery.

Habit scaffolding borrowed from streaming interfaces lengthened reading sessions without measurable comprehension cost. Future work should test whether revealing artwork suppresses self-generated imagery.

Habit scaffolding borrowed from streaming interfaces lengthened reading sessions without measurable comprehension cost. Future work should test whether revealing artwork suppresses self-generated imagery.

Habit scaffolding borrowed from streaming interfaces lengthened reading sessions without measurable comprehension cost. Future work should test whether revealing artwork suppresses self-generated imagery.

Habit scaffolding borrowed from streaming interfaces lengthened reading sessions without measurable comprehension cost. Future work should test whether revealing artwork suppresses self-generated imagery.

Fig. 1. Mean active session length by study week. Error bars show one standard deviation.

$$L(t) = a \log(1 + bt) + e, \quad e \sim N(0, s^2) \qquad (1)$$

Week	N	Mean	SD
1	20	11.2	3.4
2	20	13.8	3.9
3	19	15.1	4.2

REFERENCES

[1] A. Paivio, Mental Representations: A Dual Coding Approach. Oxford, U.K.: Oxford Univ. Press, 1986.
[2] B. J. Fogg, "A behavior model for persuasive design," in Proc. 4th Int. Conf. Persuasive Technol., 2009, pp. 1-7.
[3] M. Mangen and A. van der Weel, "The evolution of reading in the age of digitisation," J. Res. Reading, vol. 39, no. 1, pp. 116-124, 2016.
[4] K. Thompson and R. Patel, "Micro-sessions and retention in digital reading," IEEE Trans. Learn. Technol., vol. 12, no. 2, pp. 201-213, 2019.